

Rahul Raut

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Education

Sardar Patel Institute of Technology, Mumbai

Bachelor of Technology in Computer Science (Data Science)

Nov 2022 – Jun 2026

Mumbai, Maharashtra

Experience

Bluestock Fintech

Software Development Intern (Co-Team Lead)

Apr 2025 – May 2025

Remote 

- Co-led a team of 10+ developers to build a production-ready IPO web application for Bluestock
- Developed responsive user interfaces using React.js and TailwindCSS, integrating APIs for seamless workflows.
- Implemented interactive features such as IPO application forms, search and filter options, and dynamic dashboards using reusable components.

Projects

LegalEase – AI-Powered Legal Platform *React, Node.js, Express, LangChain, WebRTC* 

- Developed a full-stack legal management platform to handle cases, appointments, contracts, and blogs with complete CRUD functionality.
- Implemented automated contract generation with lawyer approval workflow and instant PDF export for verified legal documents.
- Built a RAG-based legal AI chatbot using LangChain and DeepSeekR1 for real-time PDF querying and explanation of legal terms.
- Integrated Stripe payment gateway for secure and seamless client-lawyer appointment transactions.
- Enabled real-time lawyer-client meetings using WebRTC-based video calling post successful booking and payment.

CleanCoast *React, Node.js, Express, MongoDB, WebSockets, Google OAuth, Nodemailer* 

- Created a full-stack platform allowing users to discover, register, and participate in local beach cleanup initiatives.
- Integrated Google OAuth 2.0 for secure authentication and improved onboarding experience.
- Developed a real-time group chat system using WebSockets (Socket.io) for volunteer collaboration and coordination.
- Implemented a geolocation-based email notification system to automatically alert users about nearby cleanup events.

Aerosol Classification using Hybrid ML Model *Python, Scikit-learn*

- Developed a hybrid ML pipeline combining Kernel Density Estimation, Mie scattering models, and Random Forest for aerosol classification.
- Curated an aerosol optical properties dataset with CRI, SSA, AOD, and EAE parameters across multiple wavelengths for better accuracy.
- Achieved 89% accuracy and 95% precision while training and validating on data from 47 AERONET sites globally.
- Enabled global aerosol-type mapping and pollution source analysis using a scientifically interpretable ML framework.

Technical Skills

Programming Languages: Java, Python, JavaScript

Frontend: HTML, CSS, JavaScript, React.js, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, JWT

Database: SQL, MongoDB

AI/ML: NumPy, Pandas, Scikit-learn

Data Analytics: Excel, PowerBI

Version Control: Git, GitHub

Soft Skills: Teamwork, Problem Solving, Time Management

Achievements

- LeetCode: Solved 400+ problems on data structures and algorithms, enhancing coding and analytical skills.
- Secured Top 10 position in SE Hackathon (college-level) at SPIT for innovative project development.
- Top 5 Finalist at RGIT Mumbai National Hackathon among 200+ participating teams.
- Cleared Engineering Entrance Exam JEE Advanced 2022.
- Achieved 98.5 percentile in MHT-CET with strong performance in core subjects.

Extracurricular

- Teaching Assistant – Internship Coordinator at college.
- Freelance AI Contributor at Outlier by Scale AI.